AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 32 · STUDENT LESSON

Proxy Features and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Trace the origin. Test the behavior. Keep the context.

Investigate proxy information without mistaking one deletion or equal accuracy for a complete fairness result.

Keep one habit: Complete association table → pictured mapping → full paired ablation → utility/recall comparison → independent new claim.

MAIN LESSON · SECTION 1

Which retained input still carries related information?

> A missing direct column does not establish independence or fairness.

Learning goals

I can identify possible **proxy feature**s.

I can explain why removing one field does not guarantee **fairness**.

I can propose a contextual review with concrete **evidence**.

Recall the limits of equal overall scores

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.
- Does equal overall accuracy prove fair treatment?

End goal

By the end, I can explain **proxy feature**s and **fairness** using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A feature may indirectly encode sensitive identity or context, creating unjustified differences even when the original field is absent.
</details>



LEARNING TARGET A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Trace feature origins and context before judging a proxy

> An association can create a risk without proving harm or a particular identity.

Proxy explanation and fictional-only scope

An **input** can carry information about a characteristic without naming it. In a fictional school-device dataset, device code may be associated with building or access to newer equipment.

Removing a building field does not remove all related information. Do not infer a real student identity or protected characteristic in this exercise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **proxy feature**s and **fairness**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **proxy feature**s and **fairness** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Read the association and its counterexamples

> A field can be informative without being a perfect identifier.

Complete association counts

Complete fictional association table	DeviceX	Other device	Total
North building	8	2	10
South building	2	8	10
Total	10	10	20

Fictional records show device code X on 8 of 10 north-building examples and 2 of 10 south-building examples. Device and building are associated in this sample.

This does not prove device causes model errors or that every X device comes from north. The remaining cases contradict such a universal rule.

Explain what follows and what does not

The table supports P(X|north)=8/10 and P(X|south)=2/10in this supplied sample. It does not say every Xdevice is north:2Xrecords are south. Removing building leaves device-related association, not a proved causal model effect. These building/device records do not authorize inference of a real learner's identity or protected characteristic.

The separate executable plant-card ablation below supplies actual toy prediction behavior and its own data.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

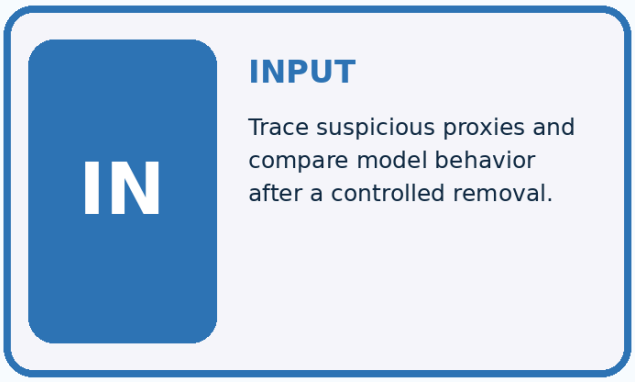
Keep the distinction between an observed result and a proposed next step.

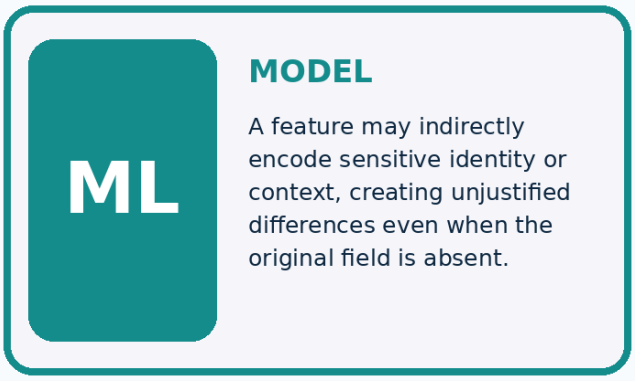
▼ **Earlier wording, alternatives and picture transcripts**

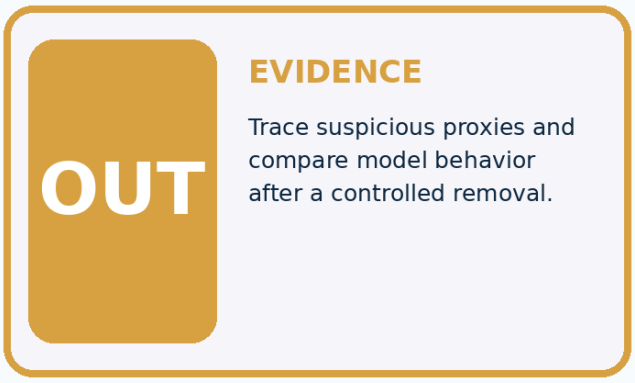
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Trace suspicious proxies and compare model behavior after a controlled removal.

MODEL IDEA A feature may indirectly encode sensitive identity or context, creating unjustified differences even when the original field is absent.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Photo background predicts classroom location and is correlated with camera quality. Model uses background as a shortcut for health labels.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

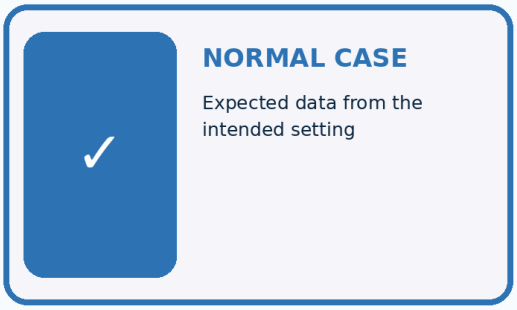
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

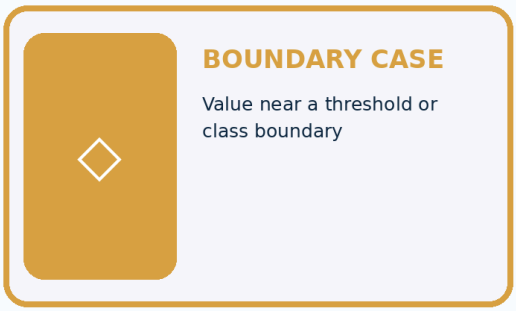
Compare dropping a direct field with removing all related information

> Actual task relevance,consequences and model behavior still need evidence.

Four evaluation conditions



NORMAL Expected data and expected behavior.

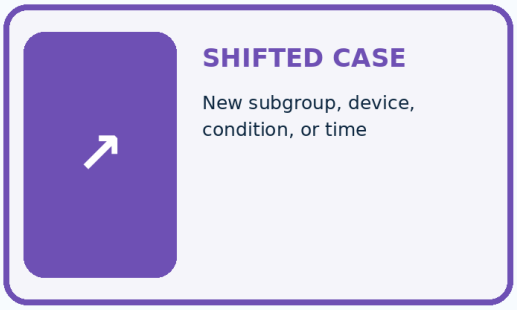


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Camera codes are assigned independently of setting in a new balanced dataset.

▼ **Check after predicting**

Prediction: camera_code no longer perfectly identifies setting in that dataset.

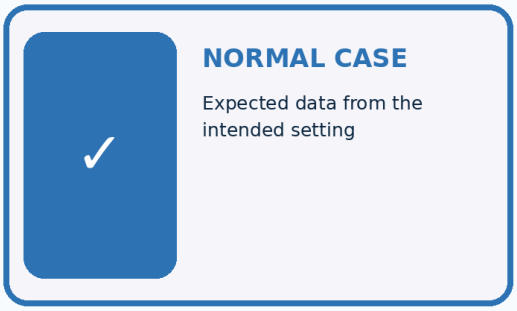
Reason: The observed mapping changed; other retained fields may still contain setting information.

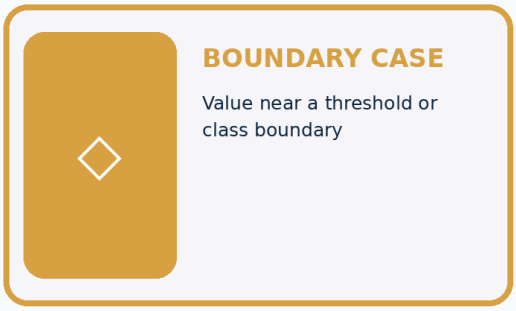
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

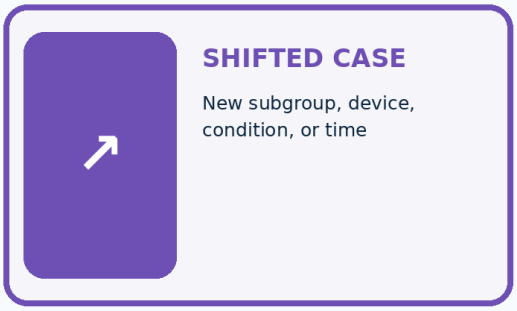
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace retained information after a direct field is removed

> Keep the original dataset and changed association record separate.

Complete pictured mapping and separate-fixture scope

Removing an obvious setting field may leave another field that carries the same information.

Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk. The team drops location.

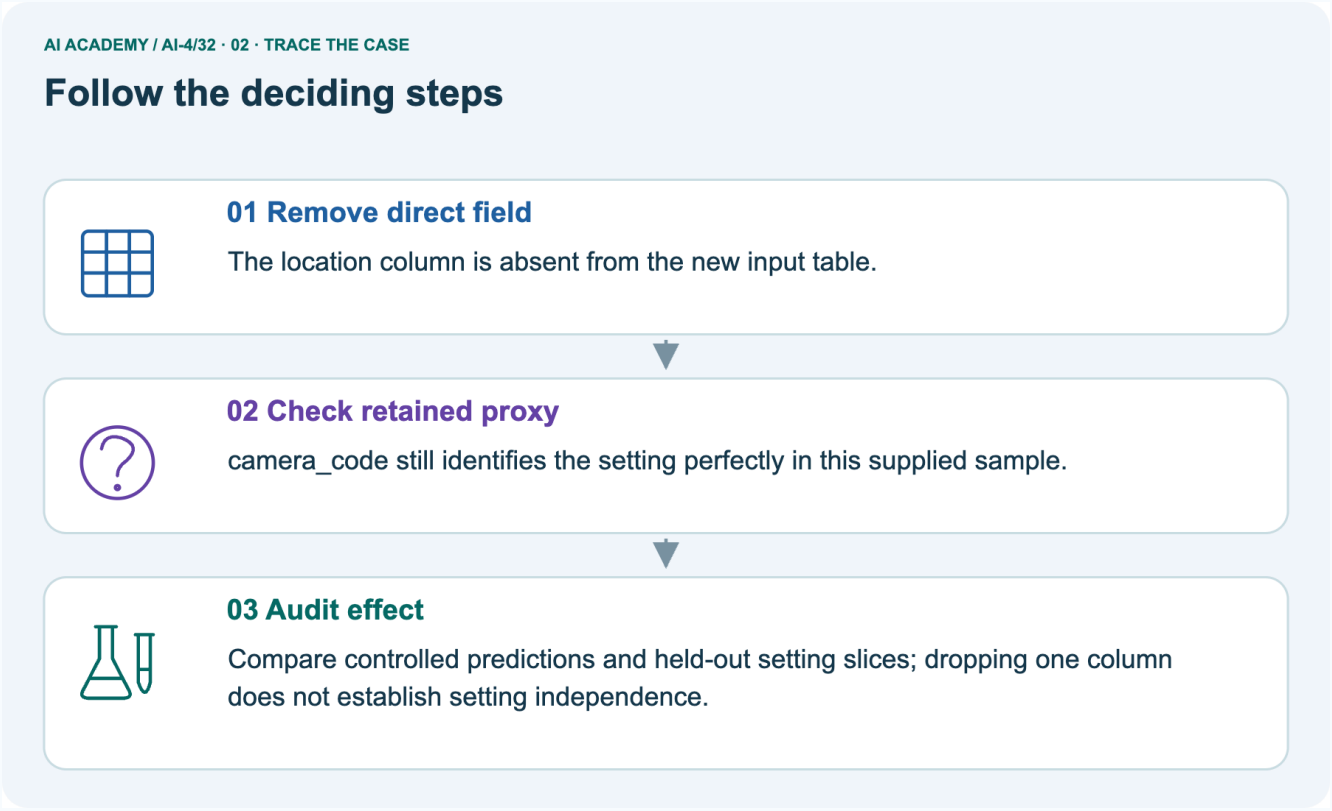
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture's original toy mapping is perfect:cameraGidentifies greenhouse and cameraDdesk in that supplied dataset, even after location is dropped. The main North/Southassociation is only 8/10 versus2/10, not perfect. A new balanced crossed

camera/setting dataset breaks the old one-to-one mapping, but that does not prove every other retained field is independent of setting.

The complete ablation trial below intentionally retains room_hintas another setting proxy; its constructed marker/label relation is not a real diagnostic or fairness result.

Trace direct-field removal,retained proxy and audit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Remove direct field	The location column is absent from the new input table.
2. Check retained proxy	camera_code still identifies the setting perfectly in this supplied sample.
3. Audit effect	Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

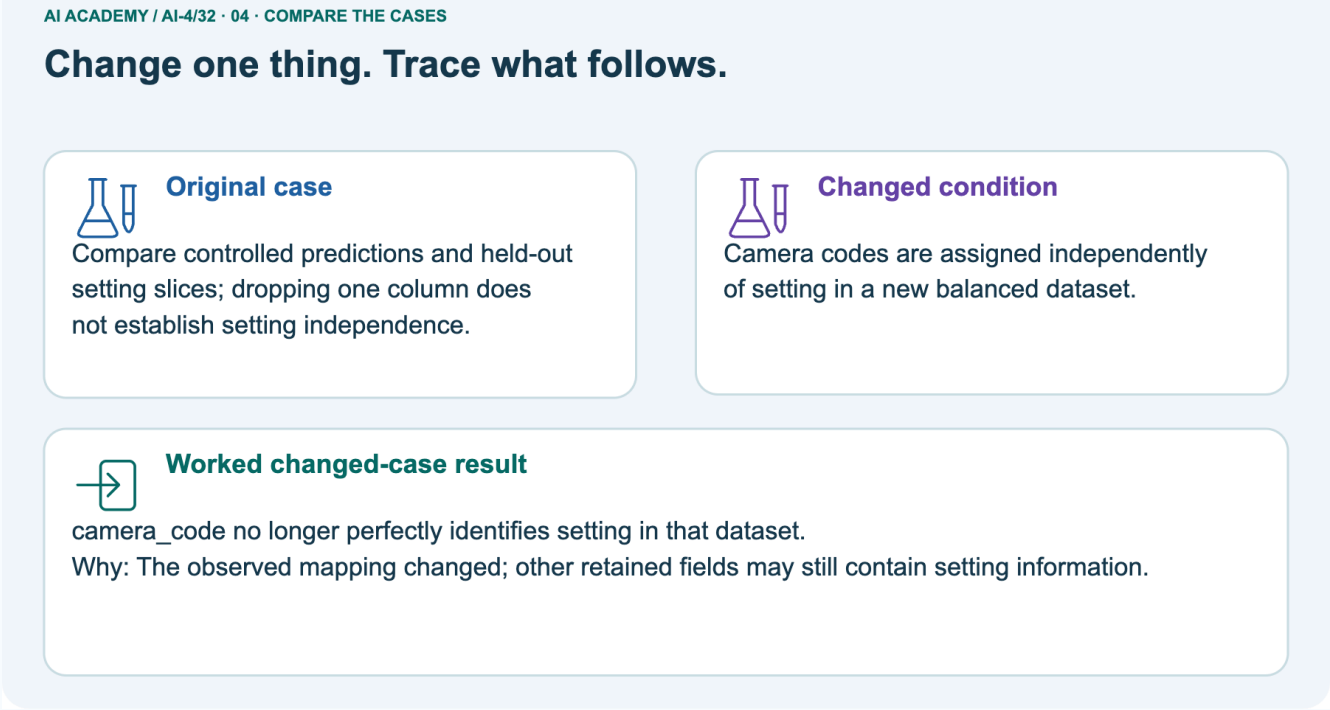
Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict crossed camera codes in a new balanced dataset

- Change one thing: Camera codes are assigned independently of setting in a new balanced dataset. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain task relevance and remaining-proxy limits

- A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

- Read the labelled picture: What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main indirect-information question

Proxy features and fairness at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
North examples	8 X of 10	2 other devices
South examples	2 X of 10	8 other devices
Review question	What does code represent?	Check context and errors

A feature may be legitimate for one purpose and inappropriate for another. Ask what is measured, when it is available, why it predicts and who might be disadvantaged. A correlation alone neither authorizes use nor proves harm. Talk through the intended decision and available alternatives.

Why might removing building but keeping device code leave related information? Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 32 • WORKED EVIDENCE

1. Photo background predicts classroom location and is correlated with camera quality.
2. Model uses background as a shortcut for health labels.
3. Create paired crops with the same leaf and different backgrounds; predictions change.
4. Crop backgrounds or improve data, retrain, and compare performance and subgroup gaps.

STEP 1 Photo background predicts classroom location and is correlated with camera quality.

STEP 2 Model uses background as a shortcut for health labels.

STEP 3 Create paired crops with the same leaf and different backgrounds; **prediction**s change.

STEP 4 Crop backgrounds or improve data, retrain, and compare performance and subgroup gaps.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

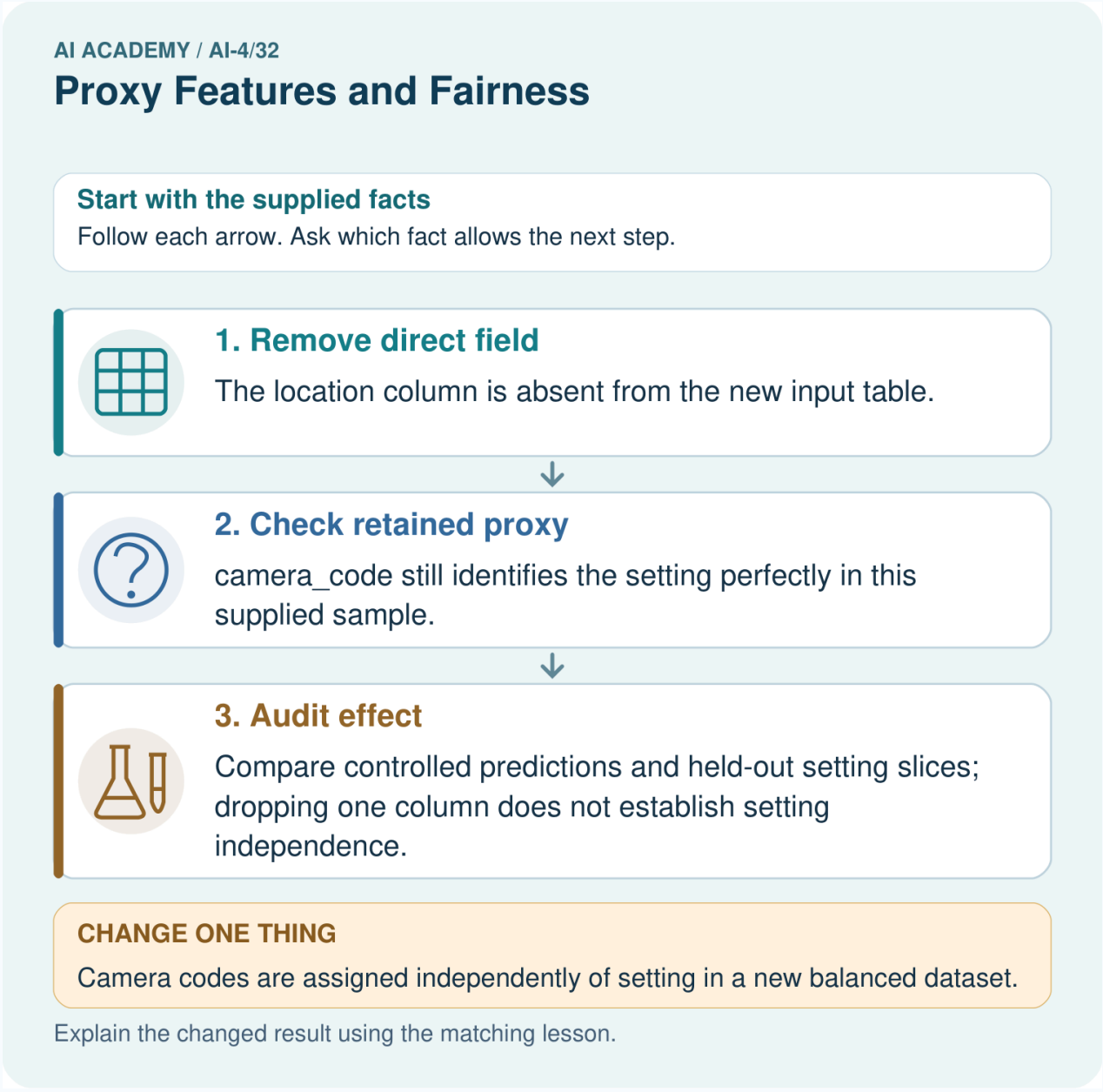
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Proxy Features and Fairness

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find one field deletion used as a fairness guarantee

> Ask which information remains and which consequences were actually measured.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Removing a sensitive column guarantees the model cannot use related information.

Explain evidence and next-test limits

- Do not advertise **fairness** through unawareness merely because names or a sensitive field were removed. A classroom audit should state unresolved questions, intended non-use and a human review path.

  Never collect sensitive student information simply to fill this exercise.

> **Does unchanged accuracy after removing a proxy prove fairness?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Removing a sensitive column guarantees the model cannot use related information. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise a controlled ablation with less help

> Compare the same cases and error types,not only a new overall score.

Try supported proxy-risk W2

Use workbook W2 for the connected example “**Proxy feature**s and **fairness**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Record ablation versions,groups and retained correlations

Compare versions with and without device code on the same validation cases, while checking subgroup errors and useful task performance.

This **ablation** measures a change in that experiment; correlated **input**s may still carry similar information. Retain group definitions and denominators, and avoid using the final test for repeated feature decisions.

Explain the evidence behind the claim

- Use the worked case for **Proxy feature**s and **fairness**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured retained setting information

- What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent postcode/school-code task

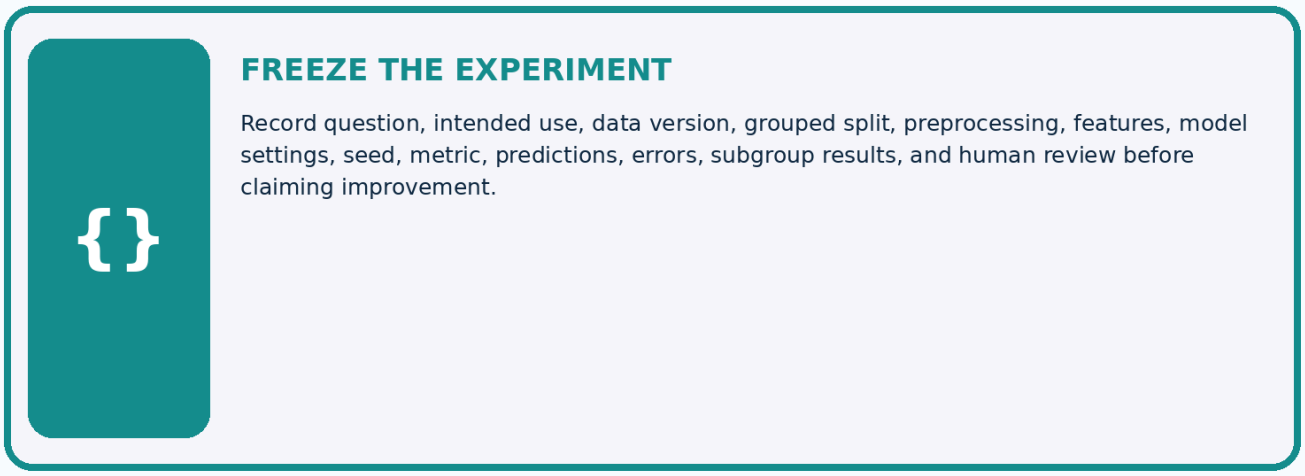
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Find a proxy risk

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete paired-test and proxy-ablation exercise

> Every input origin,training row and old/new prediction needed for the comparison is supplied.

Complete feature-origin and learner contract

Toy feature or audit field	Complete origin and role
setting	Greenhouse/desk audit metadata; not a model candidate in either version.
camera_code	BinaryG=1/D=0; perfectly tracks setting and label in the four original training records.
room_hint	Binarygreenhouse1/desk 0; another retained setting-related field.
marker	Available synthetic leaf-marker bit; perfectly matches the toy spotted label by construction. Not a real disease feature.
candidate/tie rule	V1 camera_code, room_hint, marker; V2 removes camera_code. Predict positive iff chosen bit 1; minimize training mistakes, first candidate wins ties.

Complete original training rows

Training plant · Setting · Camera bit · Room hint · Marker · Toy positive label

> **Training plant:** T1

Setting: greenhouse

Camera bit: 1

Room hint: 1

Marker: 1

Toy positive label: 1

Training plant: T2

Setting: greenhouse

Camera bit: 1

Room hint: 1

Marker: 1

Toy positive label: 1

Training plant: T3

Setting: desk

Camera bit: 0

Room hint: 0

Marker: 0

Toy positive label: 0

Training plant: T4

Setting: desk

Camera bit: 0

Room hint: 0

Marker: 0

Toy positive label: 0

Complete eight paired validation views

Validation view / plant · Setting · Camera · Room hint · Marker / reference · V1 prediction · V2 prediction

> **Validation view / plant:** P1-greenhouse-1
> **Setting:** greenhouse
> **Camera:** 1
> **Room hint:** 1
> **Marker / reference:** 1 /1
> **V1 prediction:** 1
> **V2 prediction:** 1

> **Validation view / plant:** P1-greenhouse-0
> **Setting:** greenhouse
> **Camera:** 0
> **Room hint:** 1
> **Marker / reference:** 1 /1
> **V1 prediction:** 0
> **V2 prediction:** 1

> **Validation view / plant:** P1-desk-1
> **Setting:** desk
> **Camera:** 1
> **Room hint:** 0
> **Marker / reference:** 1 /1
> **V1 prediction:** 1
> **V2 prediction:** 0

> **Validation view / plant:** P1-desk-0

Setting: desk

Camera: 0

Room hint: 0

Marker / reference: 1 /1

V1 prediction: 0

V2 prediction: 0

Validation view / plant: P2-greenhouse-1

Setting: greenhouse

Camera: 1

Room hint: 1

Marker / reference: 0 /0

V1 prediction: 1

V2 prediction: 1

Validation view / plant: P2-greenhouse-0

Setting: greenhouse

Camera: 0

Room hint: 1

Marker / reference: 0 /0

V1 prediction: 0

V2 prediction: 1

Validation view / plant: P2-desk-1

Setting: desk

Camera: 1

Room hint: 0

Marker / reference: 0 /0

V1 prediction: 1

V2 prediction: 0

> **Validation view / plant:** P2-desk-0
>
> **Setting:** desk
>
> **Camera:** 0
>
> **Room hint:** 0
>
> **Marker / reference:** 0 /0
>
> **V1 prediction:** 0
>
> **V2 prediction:** 0

Derive,ablate,compare and bound the claim

1. Proxy Detective: map the field origins, derive the tied training fits, and compare all eight paired validation outputs before running the optional complete program. State four views per plant, two new plants, and the setting-specific recall denominators. Remove only camera_codewith the other learner/data rules fixed, and explain why room_hintstill matters.

   Record missing real-world context, consequence definitions and other potential proxies rather than declaring a universal fairness score. Keep the selected postcode/school-code scenario for an uncoached first evidence request.

Checked behavior and subgroup results

Fit each version on the same four original training plants. All candidate training errors are 0, so V1 chooses camera_codeand V2 chooses retained room_hint. The eight validation views cross both cameras with both settings for each of two new plants P1/P2; related views stay together outside training. Overall correctness remains 4/8 for both versions.

V1 recall is1/2in each setting; V2 recall is2/2in greenhouse and 0/2at desk. Same overall accuracy does not imply identical errors or a fairness conclusion. The toy learner is fully specified; the marker’s perfect relation is constructed, not a real model guarantee.

Normal,paired,single-ablation and missing-context probes

Complete proxy probe	Expected bounded check
Normal main North/Southtable	DeviceX80%/20%, with counterexamples; association does not prove causal errors.
Paired toy P1 greenhouse views, camera 1 then0, room/marker held 1	V1 changes positive→negative; V2 stayspositive because it selects room_hint. This is a specified-input diagnostic, not a real observed human outcome.
Single ablation camera removed	V2 still uses setting-related room_hint; overall 4/8 does not certify independence.
Missing feature provenance or actual consequence evidence	Request/record the gap; do not invent identity, harm or fair-treatment proof.

Optional complete constrained ablation program

```
# All values are synthetic. setting is audit metadata, absent from candidates.
training = []
for item_id, bit in [('T1', 1), ('T2', 1), ('T3', 0), ('T4', 0)]:
    training.append({'id': item_id, 'setting': 'greenhouse' if bit else 'desk',
                     'camera_code': bit, 'room_hint': bit, 'marker': bit, 'actual': bit})
validation = []
for plant, marker in [('P1', 1), ('P2', 0)]:
    for setting, room in [('greenhouse', 1), ('desk', 0)]:
        for camera in [1, 0]:
            validation.append({'id': f'{plant}-{setting}-{camera}', 'plant': plant,
                               'setting': setting, 'camera_code': camera,
                               'room_hint': room, 'marker': marker, 'actual': marker})


def fit_feature(rows, candidates):
    if not rows or not candidates:
        raise ValueError('Need training records and declared candidate inputs')
    errors = [sum(r[name] != r['actual'] for r in rows) for name in candidates]
    winner = min(range(len(candidates)), key=lambda i: errors[i])
    return candidates[winner], errors
```

```
def report(rows, predictions):
    if len(rows) != len(predictions):
        raise ValueError('Need same-case paired predictions')
    correct = sum(r['actual'] == p for r, p in zip(rows, predictions))
    positives = sum(r['actual'] == 1 for r in rows)
    found = sum(r['actual'] == p == 1 for r, p in zip(rows, predictions))
    return correct, len(rows), found, positives


for version, candidates in [('V1', ['camera_code', 'room_hint', 'marker']),
                            ('V2-camera-removed', ['room_hint', 'marker'])]:
    feature, errors = fit_feature(training, candidates)
    predictions = [r[feature] for r in validation]
    print(version, feature, errors, report(validation, predictions))
    for setting in ['greenhouse', 'desk']:
        subset = [r for r in validation if r['setting'] == setting]
        print(setting, report(subset, [r[feature] for r in subset]))
# Fixed rule: predict positive iff chosen binary input is1; first candidate wins ties.
# V2 removes camera but still selects room_hint, a setting proxy in this toy sample.
# Two new plant identities/four views each; no model of real fairness or disease.
```

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Proxy Detective: map feature origins, run paired tests, remove one suspicious feature, and check both utility and gaps.

1. Write the hypothesis and expected **evidence** before running.

2. Proxy Detective: map feature origins, run **paired test**s, remove one suspicious feature, and check both utility and gaps.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: evaluate a new retained-information claim

> Request the evidence needed rather than inventing a proxy mapping.

Complete independent scenario and scope

A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

The selected postcode/school-code task gives a claimed removal, not actual association counts, model predictions or a mapping. Do not infer any real student's location or identity. State the unsupported part, request provenance/association/context evidence, and propose matched ablations and group error checks. No automatic fairness or complete-removal result follows without those observations.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 32 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A plant classifier uses camera_model, and one species was photographed mostly on a single camera. What shortcut might it learn and how could you investigate?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Document Plant Classifier proxy risks, paired-test **evidence**, **mitigation**, and remaining **fairness** limitation.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep proxy risks and unresolved context in the project record

> A mitigation claim should name what changed and what remains untested.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Document Plant Classifier proxy risks, paired-test **evidence**, **mitigation**, and remaining **fairness** limitation.

Document possible proxies, their origin, task relevance and time availability, and the actual evidence for model sensitivity. Compare matched versions and subgroup/error types with source counts; retain other correlated fields and unresolved risks.

Fairness requires context about affected parties, consequences and appropriate evaluation goals; neither removing a column nor matching overall accuracy supplies that context. This course’s object-only toy examples establish no real-person fairness conclusion.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

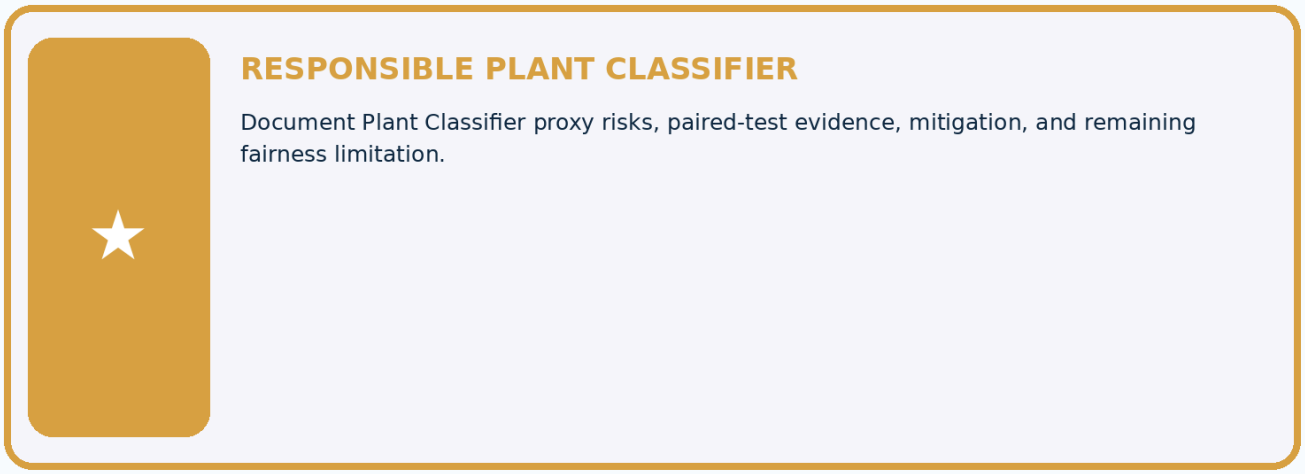
My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish with the retained information and the evidence still needed

> Explain why association,model sensitivity,harm and fairness are distinct claims.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Identify possible **proxy feature**s

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
proxy feature	input correlated with another sensitive attribute	
fairness	evaluate benefits and harms across affected groups	
paired test	cases differing in one relevant factor	
mitigation	control reducing an identified risk	

Closer-explanation terms

Word	Meaning
Proxy feature	A variable that can stand in for another characteristic
Correlation	An observed association, not proof of cause
Sensitive attribute	A characteristic requiring particular care in a context

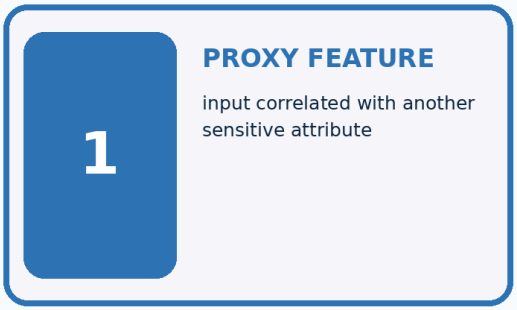
Word	Meaning
Ablation	A comparison that removes a feature or component

▼ Earlier wording, alternatives and picture transcripts

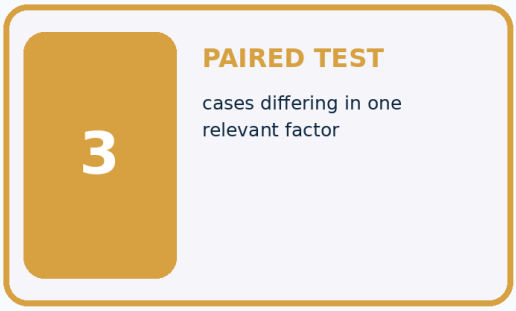
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Removing a sensitive column guarantees the model cannot use related information.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **proxy feature**, **fairness**, **paired test**, **mitigation**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Removing a sensitive column guarantees the model cannot use related information.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

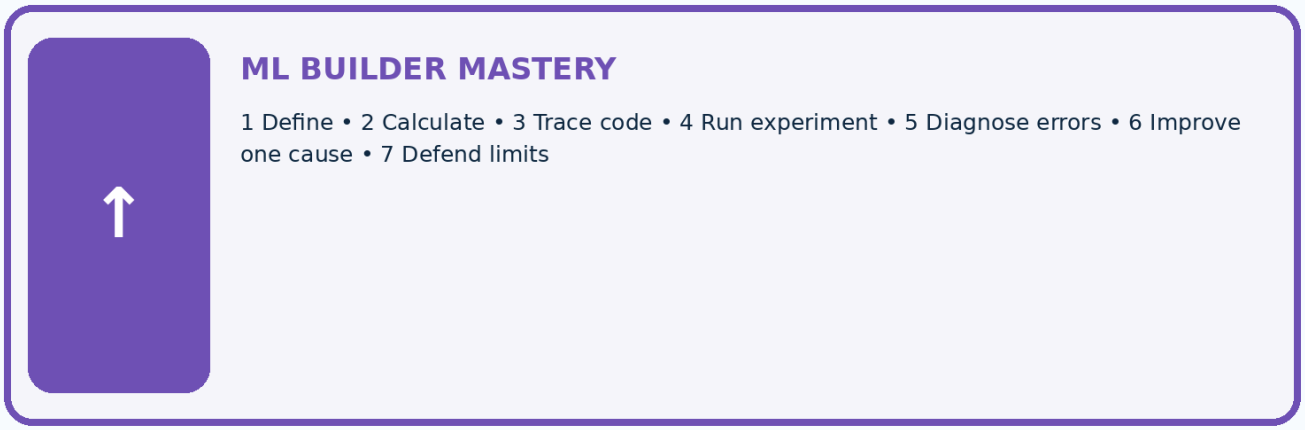
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh postcode/school-code evidence-request task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, **fairness** concern, and the human-review action.

- Independent **evidence** and defense:

- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Trace suspicious proxies and compare model behavior after a controlled removal.

NEXT BRIDGE Carry the experiment **evidence** into: Model Cards and Human Review.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Background colour tracks collection location, associated with plant types.

Explain proxy risk and test.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

Evidence or stage	Value or result
Background colour	Tracks collection location
Location	Associated with plant type
Risk	Model uses shortcut
Test	Cross leaf types with backgrounds and locations

▼ Earlier wording, alternatives and picture transcripts

Week 32 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Test whether a proxy replaces leaf evidence

Background colour tracks collection location, associated with plant types.

Explain proxy risk and test.

Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Identify possible proxy features	Explain the deciding step, show a trace or test, and state one limit.
Explain why removing one field does not guarantee fairness	Explain the deciding step, show a trace or test, and state one limit.
Propose a contextual review with concrete evidence	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Removing an obvious setting field may leave another field that carries the same information.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Removing an obvious setting field may leave another field that carries the same information.

- Which labelled step matches this detail? Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.
- 1. Remove direct field
- 2. Audit effect
- 3. Check retained proxy
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Audit effect Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence. Read the picture in order.

Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Camera codes are assigned independently of setting in a new balanced dataset. Which conclusion is supported by the supplied facts?

- 1. camera_code no longer perfectly identifies setting in that dataset.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: camera_code no longer perfectly identifies setting in that dataset. The observed mapping changed; other retained fields may still contain setting information.

Repair and keep the record

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

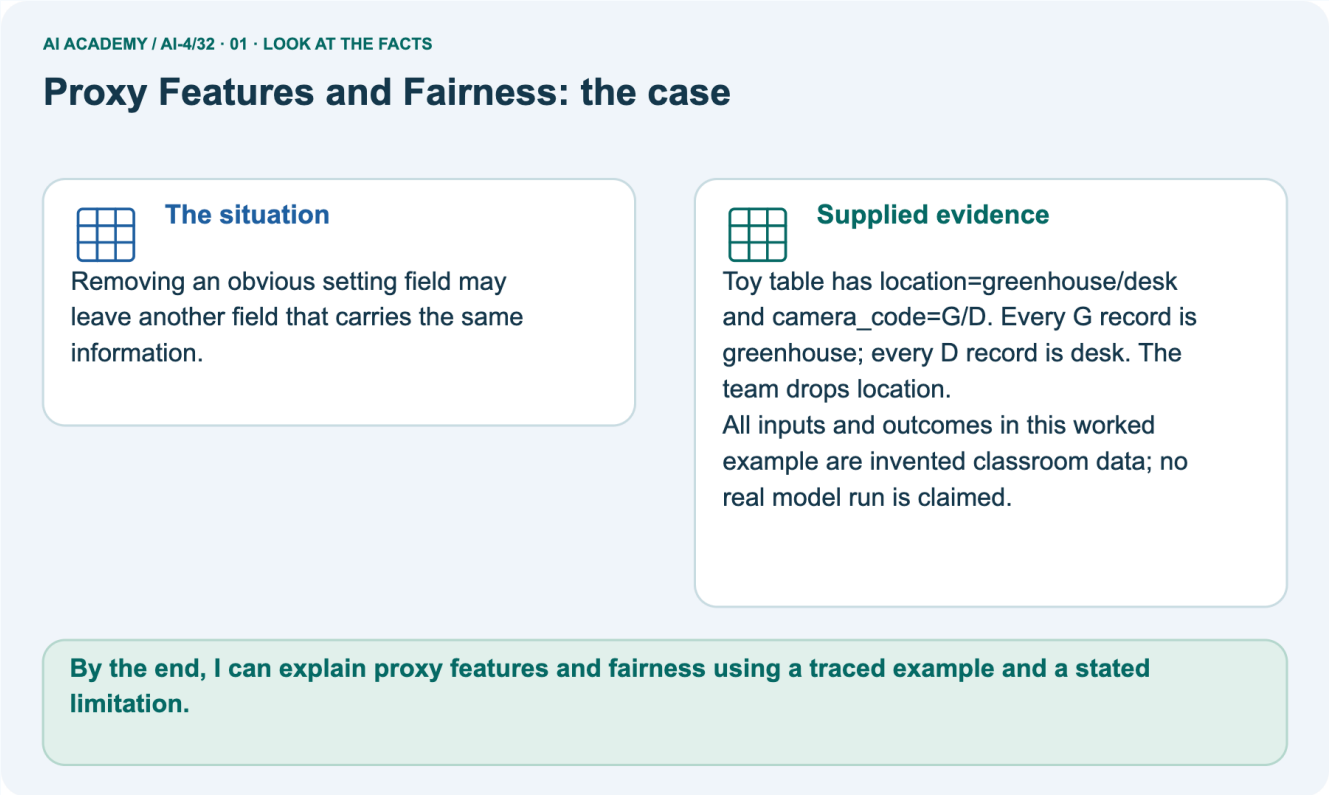
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Proxy Features and Fairness: the case



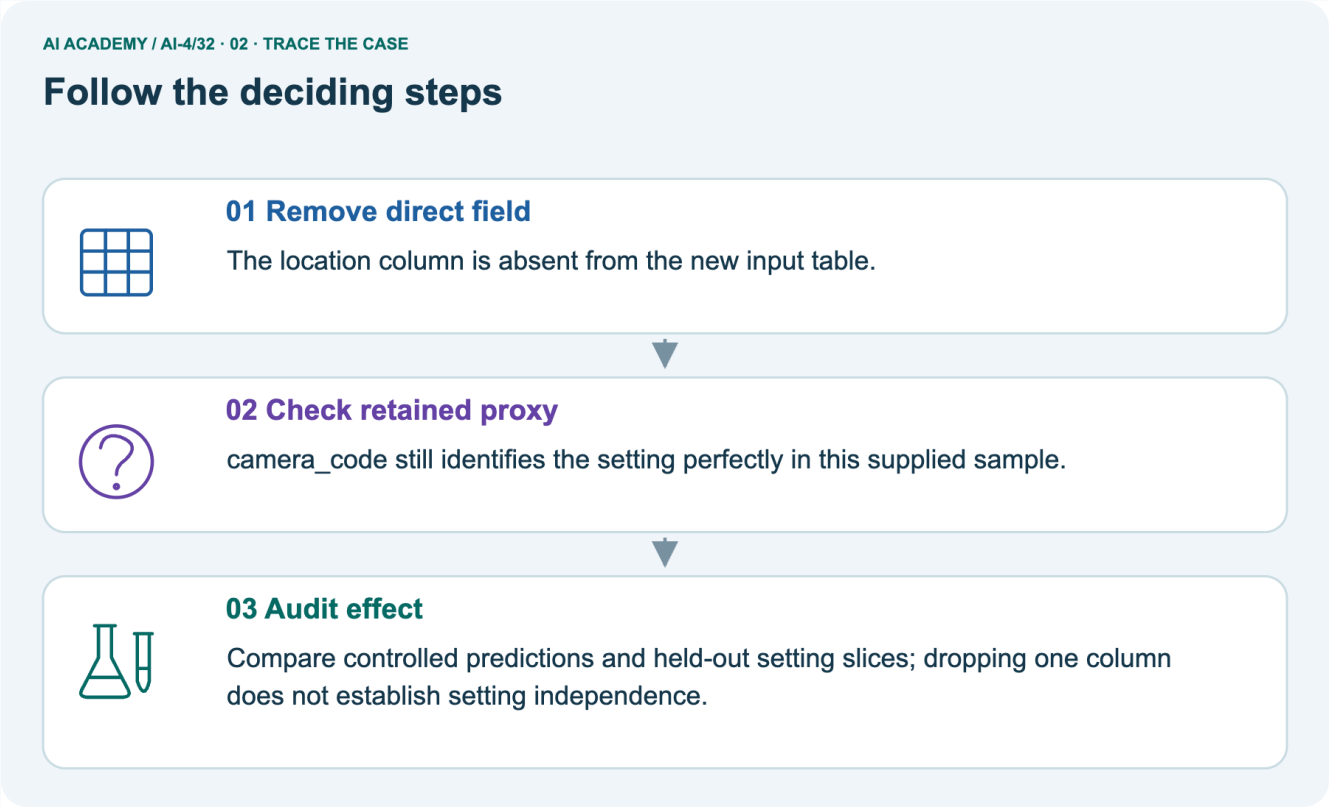
Follow the labelled picture with the matching explanation. Open the vector diagram

Removing an obvious setting field may leave another field that carries the same information. Supplied facts: Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk.

The team drops location. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

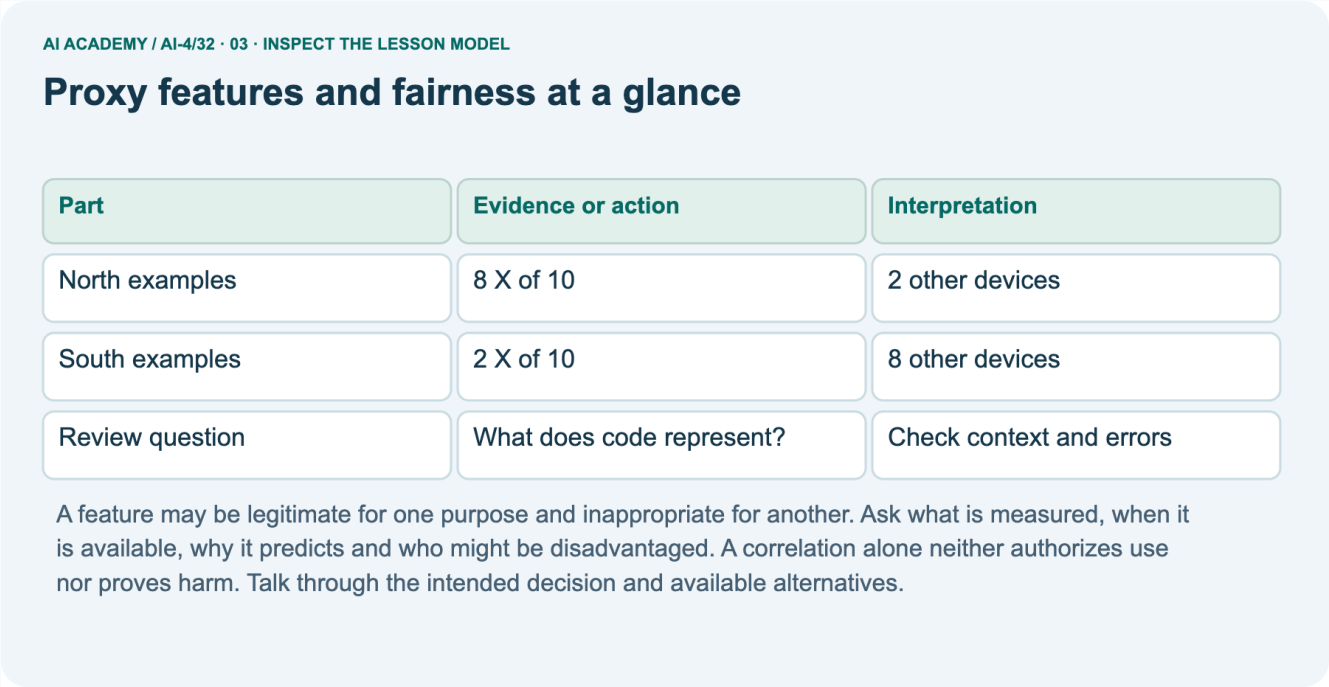


Follow the labelled picture with the matching explanation. Open the vector diagram

Remove direct field: The location column is absent from the new **input** table. → Check retained proxy: camera_code still identifies the setting perfectly in this supplied sample. → Audit effect: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Point to each numbered stage and explain which recorded **input** it uses.

03. Proxy features and fairness at a glance

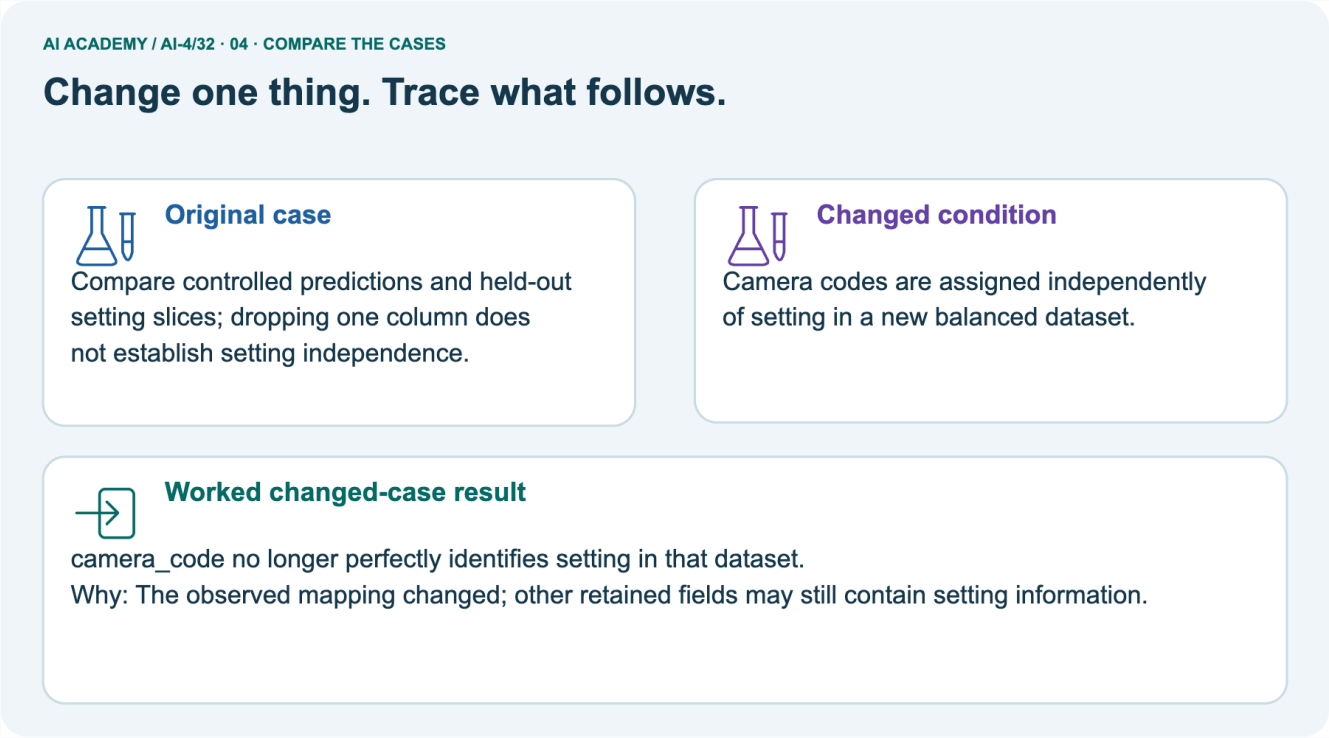


Follow the labelled picture with the matching explanation. Open the vector diagram

Proxy features and **fairness** at a glance. North examples | 8 X of 10 | 2 other devices. South examples | 2 X of 10 | 8 other devices. Review question | What does code represent? | Check context and errors

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

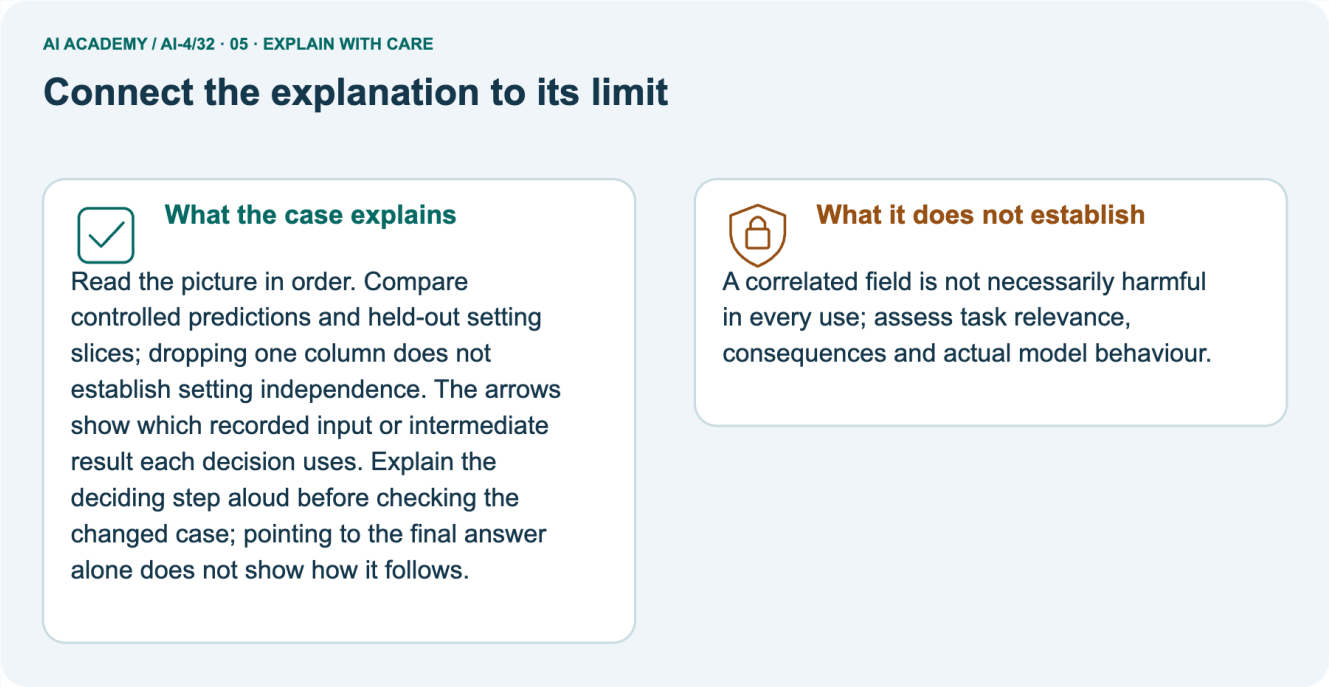
Original case: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Changed condition: Camera codes are assigned independently of setting in a new balanced dataset. Expected result: camera_code no longer perfectly identifies setting in that dataset.

Reason: The observed mapping changed; other retained fields may still contain setting information.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

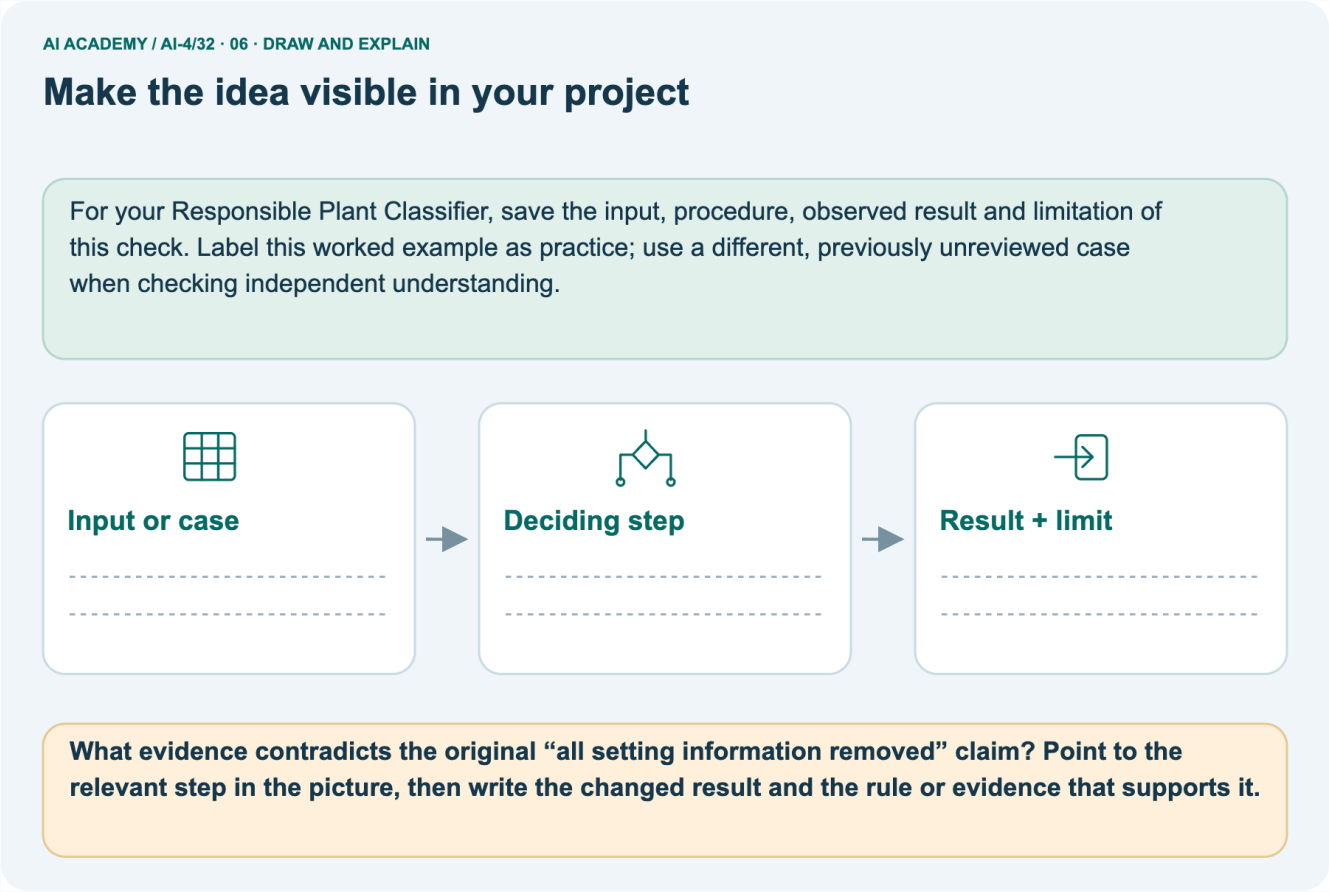
Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Proxy Features and Fairness: the case

02. Follow the deciding steps

03. Proxy features and fairness at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.